

*Software Engineer with 5+ years experience designing scalable microservices, distributed systems, and cloud-native applications. Experienced in **integrating AI/ML features** into production systems. Strong foundation in **Python, C#, React, REST and gRPC**, and cloud platforms with a proven ability to deliver production-ready solutions. I enjoy solving complex problems and building systems that are simple, scalable and maintainable.*

TECHNICAL SKILLS

- **Programming Languages:** Python, C#, Java, JavaScript, SQL
- **Web & Frontend:** React, Redux
- **Backend & Microservices:** .NET Core, Django, Flask, FastAPI, REST/gRPC APIs
- **Databases & Storage:** PostgreSQL, MySQL, MongoDB, Redis, Pinecone, FAISS, Supabase, CromaDB, Snowflake, AWS Redshift
- **Cloud & DevOps:** AWS, GCP, Docker, Git, MLflow
- **Applied AI / ML (Enhancement):** LLM integration, RAG pipelines, embeddings, prompt engineering, multimodal models, clustering (HDBSCAN, UMAP), quantization, model fine-tuning, TensorFlow, PyTorch, Numpy, Pandas, Scikit-learn

WORK EXPERIENCE

AI/ML Engineering Intern (part of MS internship course) Pyramyd.ai (May 2025 – Aug 2025), Berkeley, CA

- Architected an **AI-powered software scoring platform** for 3000+ SaaS vendors using **RAG pipelines, model fine tuning and multi agent orchestrations** achieving 300 ms latency and 99.9% uptime
- Implemented **MCP-based AI agents** to orchestrate retrieval, scoring, and recommendation flows across LLM, graph, and sentiment models, improving context adaptation and reducing redundant calls by 25%.
- Developed **contrastive learning** pipeline improving vendor–buyer match recall by 14% and deployed it via **FAST APIs**.
- Performed **PEFT** and **LoRA** fine-tuning on OpenAI and Open source models for extracting aspects and sentiments from user reviews
- Designed a graph-based model to depict competitors & capabilities of software using feature and review similarity.
- Automated model refresh and deployment with **MLflow** on **GCP**, reducing manual model retraining by 3.3x.

Senior Software Engineer Musafir.com India Pvt. Ltd. (Jul 2018 – Nov 2023), India

- Engineered a **hybrid recommender system** to personalize travel suggestions and raise booking conversion by **25%** and click through rate by **30%**.
- Developed an **autoencoder-based fraud detection** pipeline blocking **80%** fake bookings, saving the company **USD 90,000** annually in chargebacks.
- Introduced **async operations, lazy loading, and connection pooling** in flight search APIs which reduced search time from **4sec** to **800 ms**.
- Led migration from a legacy monolith to **.NET Core microservices**, reducing API latency from **800ms** to **400ms** which improved availability and scaled the platform to **1M daily requests**.
- Built high-performance **REST** and **gRPC APIs** and event-driven pipelines using **Kafka** and **Spark**, for flight, seat and price tracking systems.
- Built a centralized **auth framework (OAuth2, JWT, SSO)** that unified login flows across 8+ web and mobile apps, improving user experience and reducing security vulnerabilities by standardizing token handling.
- Delivered a **Dialogflow chatbot** to enable bookings with chats and reduce repetitive support queries by 20%.
- Optimized **PostgreSQL, MongoDB** and **Redis** through indexing, partitioning, and **caching**, improving query performance by 45% at peak load.
- Implemented a **tagging & metadata framework**, using **.Net** and **React-Redux** UI, to classify and filter data for targeted customer use cases.
- Developed **React-based UI components** and internal finance dashboards to streamline operations and cut support tickets by 28%.
- Architected and developed an application for **multi-level approval workflows** and conversations, enabling corporate customers to get travel approvals, communicate with internal teams, and travel partners.
- Integrated more than 10 third-party airline, hotel, and supplier APIs via modular adapters, improving inventory coverage and supplier resiliency.
- Built a **hackathon-winning NLP “text-to-booking” prototype** that enabled bookings from SMS text, restoring critical revenue of **USD 100,000**.
- Led **design discussions** and **code reviews**, streamlining **sprints** and improving team delivery velocity by ensuring clean, maintainable code.
- Mentored engineers and led a **12-member agile team**, also authored onboarding guides and technical docs to shorten ramp-up time.

SELECTED PROJECTS

LLM-Based Interview Simulation with RAG & Multimodal Evaluation - GPT-4o, LLaMA-2, Mistral-7B, Qwen, ViT, Whisper, Praat, RAG, GCP, Pinecone

- Developed a RAG pipeline using text embeddings of interview questions & answers by populating in **ChromaDB** for fast, metadata-aware retrieval.
- Fine-tuned **LLaMA-3-70B (LoRA, 4-bit)** and **Mistral-7B** on question-answer-feedback samples, which resulted in fluent, adaptive interviewing.
- Integrated **OpenAI Whisper** with **Librosa** noise-reduction module for real-time speech-to-text conversion, enabling immersive voice interviews.

Customer Support Chatbot using LLMs - Groq, LLaMA-3, Supabase, Next.js, TypeScript, Vercel

- Shipped a production **support assistant** that turns a static FAQ into conversation style chat to answer user queries using RAG & LLMs using **Groq**.
- Implemented **guardrails** for uncertain answers to trigger a follow-up, and unresolved cases to become auto-generated tickets with summaries.
- Delivered a clean **chat UI** with conversation history and **authentication** framework using **Next.js, TypeScript and Supabase** and deployed on **Vercel** with environment management.

EDUCATION

- **MS in Applied Data Intelligence, San Jose State University, San Jose, California, USA,** Expected Dec 2025, GPA 3.87/4.0
Relevant Coursework: Distributed Systems, Deep Learning, Generative AI, Machine Learning, Big Data Technologies
- **B.E.(Information Technology), VIIT, Pune University, India** 2014 - 2018, GPA 3.71/4.0